

Basics of Modelling the Pedestrian Flow

IT4033E Mathematical Modeling – Group 2

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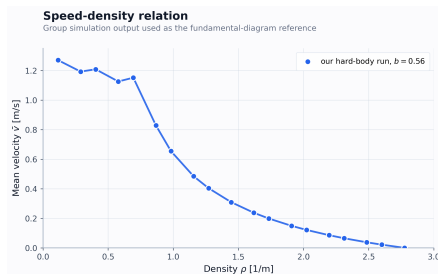
Pedestrian flow modelling links individual motion and interactions to crowd-level density, speed, and congestion, supporting real-world applications (facility design, bottleneck analysis,...)



Two model families

- *Cellular automata*
- *Continuous-space models*
⇒ Focus of this paper.

Speed–Density Relation



Group simulation output: higher density $\Rightarrow$ lower mean walking speed.

- **Fundamental diagram:** relationship between mean velocity v and density ρ reproduced by our 1D simulation.
- **Validation target:** a valid pedestrian model should reproduce the observed v – ρ curve.

Why Use a 1D Model?

Single-file flow (1D) $\approx$ Planar flow (2D)

Similar speed–density behaviour: $v(\rho)$

- **Empirical observation:** 1D and 2D pedestrian flows show similar velocity–density curves in both shape and magnitude.
- **Limited lateral effect:** lateral interactions do not noticeably change the fundamental diagram up to $\rho \approx 4.5 \text{ m}^{-2}$.
- **Therefore:** a 1D model retains the essential macroscopic behaviour while isolating the effects of required space and interaction.

Introduction: Research Objectives

- **Paper:** Seyfried, Steffen & Lippert, *Basics of Modelling the Pedestrian Flow*, Physica A 368 (2006)
- **Main goal:** modify the 1D social force model so that it reproduces the empirical fundamental diagram
- **Two specific questions driving the modification:**
 - 1 Does making the required space depend on velocity improve the fit to empirical data?
 - 2 How much does *remote action* (a smooth repulsive force acting before contact) actually matter for the resulting v - ρ curve?

Motivation: The Original Social Force Model

Social Force Model: Each pedestrian is modelled as a self-driven object in continuous space, characterised by position $x_i(t)$, velocity $v_i(t)$, and mass m_i . For pedestrian i , its dynamics are described by:

$$\frac{dx_i}{dt} = v_i, \quad m_i \frac{dv_i}{dt} = F_i = F_i^{drv} + F_i^{rep}$$

- **Driving force** F_i^{drv} : drives pedestrian i toward the intended velocity v_i^0 .

$$F_i^{drv} = m_i \frac{v_i^0 - v_i}{\tau_i}$$

- **Repulsive force** F_i^{rep} : models the tendency to maintain personal space and prevent collisions through interactions with other pedestrians $j \neq i$.

$$F_i^{rep} = \sum_{j \neq i} -\nabla A_i(\|x_j - x_i\| - d_i)^{-B_i}$$

⇒ Even in this simple form, the model can reproduce collective effects such as lane formation and oscillations at bottlenecks.

Motivation: Why Modify the Model for 1D?

In a single-file setting, the original interaction needs to be adapted:

1. Physically valid motion

A symmetric repulsion may cause backward motion ($v_i < 0$) or accelerate a pedestrian beyond their intended speed ($v_i > v_i^0$).

2. Front-only influence

In single-file walking, motion should be governed by the pedestrian directly ahead, not by pedestrians behind.

3. Speed-dependent required space

A fixed required length does not reproduce the empirical velocity–density relation correctly.

The required length should increase with walking speed:

$$d_i(t) = a_i + b_i v_i(t)$$

This captures the fact that faster pedestrians require more space in front of them.

Forward-only interaction + speed-dependent required space

⇒ The reduced 1D model isolates how required space and remote interaction shape the fundamental diagram.

Methodology: Problem Setting and Mathematical Model

- **Setup:** N pedestrians confined to a 1D corridor of length L , with periodic boundary conditions
- **Ordering convention:** pedestrians are indexed so that $i+1$ is always the one immediately ahead, i.e. $x_{i+1}(t) > x_i(t)$; all masses normalized to $m_i = 1$
- **Density:** $\rho = N/L$
- **Equation of motion for each pedestrian:**

$$\frac{dx_i}{dt} = v_i(t), \quad \frac{dv_i}{dt} = F_i(t)$$

- **Key departure from the original model:** $F_i(t)$ is no longer a superposition $\sum_{j \neq i} F_{ij}$ over every other pedestrian
 - Instead, $F_i(t)$ is determined **exclusively** by the state of pedestrian $i+1$ in front
- This single change in scope is what allows the system to be analyzed and simulated as a chain of one-sided, local interactions rather than a fully coupled many-body force field

Methodology: Modifications to the Social Force Model

- **Modification 1 : One-sided (forward-only) interaction**

- The sum $\sum_{j \neq i}$ is collapsed into a single term involving only pedestrian $i + 1$
- In single-file motion, a person's pace is dictated by the gap ahead, not by anyone behind

- **Modification 2 : Hard velocity bound**

- Resulting velocity is restricted to the physically sensible range $v_i \in [0, v_i^0]$
- Lower bound at 0: rules out backward motion induced by repulsion
- Upper bound at v_i^0 : rules out being "pushed" faster than one's own intended speed

- **Modification 3 : Required length as a function of velocity**

- The fixed hard-core size d_i is replaced by a quantity that grows with current speed:

$$d_i(t) = a_i + b_i v_i(t)$$

- Captures the empirical fact that a faster-moving pedestrian occupies more space along the direction of travel

Interaction Force Models: Hard body (no remote action)

- **Force law:**

$$F_i(t) = \begin{cases} \frac{v_i^0 - v_i(t)}{\tau_i}, & x_{i+1}(t) - x_i(t) > d_i(t) \\ -\delta(t) v_i(t), & x_{i+1}(t) - x_i(t) \leq d_i(t) \end{cases}$$

with $d_i(t) = a_i + b_i v_i(t)$

- Depends on exactly three quantities: pedestrian i 's own position and velocity, and the position x_{i+1} of the one ahead
- **Free regime** ($\text{gap} > d_i$): only the driving term acts, pedestrian accelerates toward v_i^0
- **Contact regime** ($\text{gap} \leq d_i$): velocity is forced to zero — an instantaneous switch, not a smooth decay
- Since d_i scales with v_i , a faster pedestrian needs more space
- There is no anticipation built in: the model reacts only once the threshold is actually crossed, never before

Interaction Force Models: Hard body (with remote action)

- **Force law:**

$$F_i(t) = \begin{cases} G_i(t), & v_i(t) > 0 \\ \max(0, G_i(t)), & v_i(t) \leq 0 \end{cases}$$
$$G_i(t) = \underbrace{\frac{v_i^0 - v_i(t)}{\tau_i}}_{\text{driving}} - \underbrace{e_i \left(\frac{1}{x_{i+1}(t) - x_i(t) - d_i(t)} \right)^{f_i}}_{\text{remote repulsion}}$$

- Still strictly one-sided: only the pedestrian ahead contributes to G_i
- The repulsive term grows without bound as the gap approaches $d_i(t)$, so deceleration begins **before** contact is actually made
- e_i sets the strength of the repulsion; f_i sharpness as gap narrows
- Stopped pedestrian cannot be pushed backward.
- Consequence: with e_i, f_i properly tuned, full stops happen mainly during the early relaxation phase, far less often.

Time Stepping: Hard body with Remote Action

- **Why it is simple here:** $G_i(t)$ is continuous along the trajectory
- **Scheme used: explicit Euler**, with $\Delta t = 0.001$ s:

$$v_i(t + \Delta t) = v_i(t) + \Delta t \cdot F_i(t)$$

$$x_i(t + \Delta t) = x_i(t) + \Delta t \cdot v_i(t + \Delta t)$$

- Since Δt is small relative to how fast inter-pedestrian gaps actually change, the standard Euler discretization error stays negligible

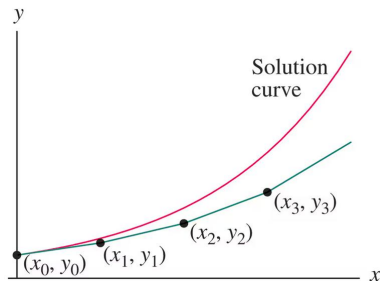


Figure: Example of Euler's method

Time Stepping: Hard body without Remote Action — The Problem

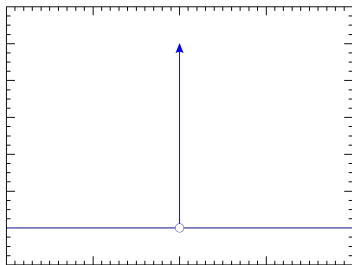


Figure: Dirac spikes at collision times t^* are unknown until the trajectory is solved

- F_i is **discontinuous**: switches abruptly between driving term and stop
- A jump in v_i at collision time t^* means $\frac{dv_i}{dt}$ contains a **Dirac spike**.
 - The RHS is a *distribution*, not a regular function - standard ODE solvers cannot handle it directly
- The exact collision time t^* is **not known in advance**.

Time Stepping: Corrective Algorithm (Steps 1–3)

Key idea: predict first with plain Euler, then fix any violations afterward

- ➊ **Predict** : update every pedestrian as if unconstrained:

$$v_i^{new} = v_i + \Delta t \cdot \frac{v_i^0 - v_i}{\tau_i}, \quad x_i^{new} = x_i + \Delta t \cdot v_i^{new}$$

- ➋ **Check** : compute spacing for every pair $(i, i+1)$:

$$s_i = x_{i+1}^{new} - x_i^{new}$$

- ➌ **Correct** : if $s_i \leq d_i$, undo pedestrian i 's move and stop them:

$$x_i^{new} \leftarrow x_i^{old}, \quad v_i^{new} \leftarrow 0$$

- Pedestrian i is projected back to its last valid position

Time Stepping: Corrective Algorithm (Steps 4–5)

4 **Propagate**: re-examine the pedestrian directly behind i

- Rolling i back to x_i^{old} shrinks the gap for pedestrian $i-1$:

$$s_{i-1} = x_i^{old} - x_{i-1}^{new} \longrightarrow \text{may now violate } s_{i-1} \leq d_{i-1}$$

- If so, apply Step 3 to $i-1$ as well, then check $i-2$, and so on

5 **Iterate** : repeat Steps 3–4 until no overlap remains:

$$s_i > d_i \quad \forall i$$

Why propagation matters

Freezing one pedestrian can trigger a *cascade* of corrections down the queue - especially under periodic boundary conditions, where the chain wraps around the entire system.

Time Stepping: Corrective Algorithm — Visual Summary

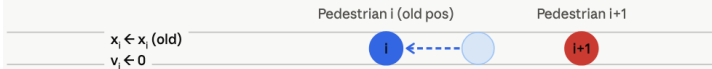
Step 1 — Predict



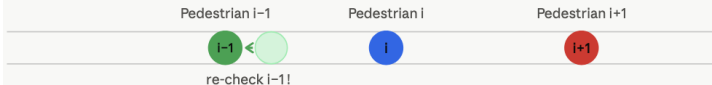
Step 2 — Check



Step 3 — Correct



Step 4 — Propagate (cascade)



Repeat until no overlap remains anywhere in the system

Simulation Setup

- **Environment:** Periodic boundary conditions, length $L = 17.3$ m.
- **Input Parameters:**
 - **Intended velocity:** $v \approx$ normal distribution with $\mu = 1.24$, $\sigma = 0.05$ m/s.
 - **Relaxation time:** $\tau = 0.61$ s.
 - **Minimum stationary space:** $a = 0.36$ m.
- **Simulation Protocol:**
 - Beginning with 3×10^5 relaxation steps.
 - Then 3×10^5 measurement steps.
- **Empirical data:** Extracted via plot digitization from Seyfried et al. (2005) reference.

Hard-body Model without Remote Action

Space-Velocity Coupling (b):

● Constant Space ($b = 0$):

- Fails to replicate empirical trends.
- **Incorrect curvature:** sharp velocity drop at low densities.

● Dynamic Space ($b > 0$):

- Restores characteristic **convex** shape.
- **Optimal fit:** $b = 0.56$ s (Seyfried data).

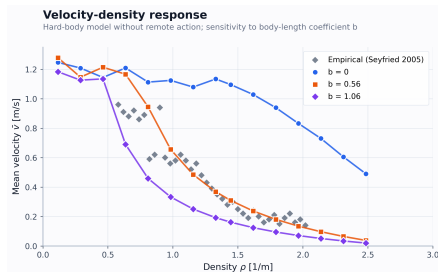


Fig 1: Impact of parameter b with Seyfried empirical points.

Key Finding

Empirical data is **only reproducible** if required space (d) scales with velocity ($b > 0$).

Hard-body Model with Remote Action

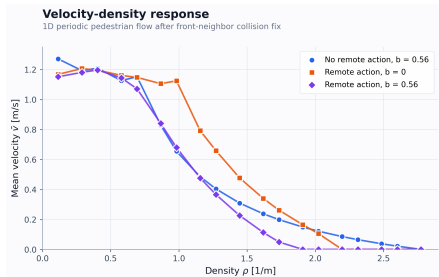


Fig 2: v - ρ curves with remote forces.

Impact of Remote Forces:

- **Dynamic Space ($b = 0.56$ s):**
 - **Negligible effect** on macro diagram.
 - Identical to model without remote action.
- **Constant Space ($b = 0$):**
 - Unphysical "**velocity gap**" at $\rho \approx 1.2 \text{ m}^{-1}$.
 - Flow drops abruptly to near-zero.

Critical Observation

Remote forces only distort macroscopic flow if spacing ignores velocity ($b = 0$).

Microscopic Mechanism: Stop-and-Go Waves

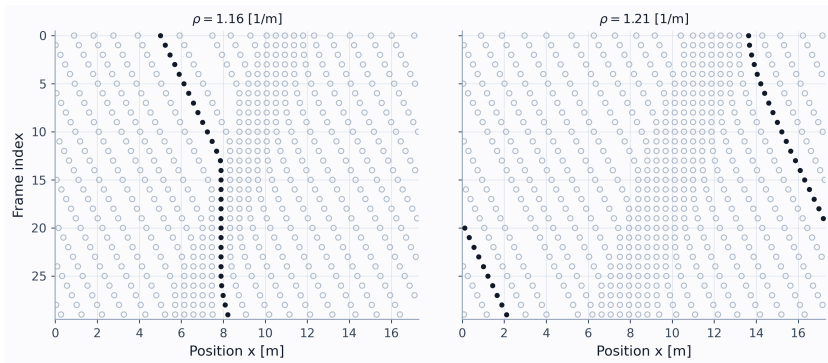


Fig 3: Same pedestrian tracked across $\rho = 1.16$ and $\rho = 1.21 \text{ m}^{-1}$ ($b = 0$).

Instability & Velocity Gap ($b = 0$):

- **Transition:** stable flow ($\rho = 1.16$) → amplified perturbations ($\rho = 1.21$).
- **Cascade:** leader slows → followers brake sharply → **backwards-propagating waves**.

Stabilization ($b > 0$)

Slower agents reduce required space $d(v)$, adding organic damping that suppresses artificial waves.

Parameter Analysis: New Sweep

- **New varied parameter:** minimum required space a in

$$d_i(t) = a + bv_i(t).$$

- Keep $b = 0.56$ s, $\tau = 0.61$ s, and $L = 17.3$ m fixed.
- Sweep $a \in \{0.30, 0.36, 0.42\}$ m.
- **Reason for this sweep:** a changes the pedestrian's baseline footprint before any velocity-dependent spacing is added.
- **Question:** how much does this static footprint shift the onset of congestion?

Parameter Analysis: Minimum Required Space (a)

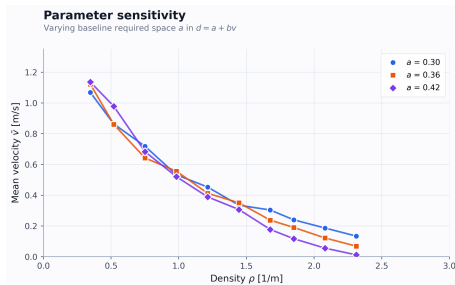


Fig 4: Fundamental diagram under different baseline spaces a .

Impact of Baseline Space:

- **Low Density ($\rho < 0.8 \text{ m}^{-1}$):**
 - Curves remain close.
 - **Reason:** pedestrians rarely hit the hard-body constraint.
- **Medium–High Density:**
 - Larger a reduces free space earlier.
 - The velocity drop shifts to lower densities.

Parameter Analysis: Interpretation

Physical Meaning of a :

- a is the non-negotiable body or personal-space baseline.
- It remains active even when pedestrians are almost stopped.
- Increasing a lowers the effective packing capacity.

Model Consequence:

- b controls how spacing grows with speed.
- a controls how early hard-body blocking starts.
- The fundamental diagram is therefore sensitive to both static and dynamic space terms.

Key Finding

Calibrating a changes congestion onset even when b is fixed at the paper's best-fit value.

Key Research Outcomes

Model Reconstruction from Scratch:

- **Directional constraints:** converted the 2D symmetric Social Force Model into a one-sided, forward-only 1D formulation.
- **Velocity bounds:** clamped velocities to $[0, v_i^0]$ to avoid backward motion and overshooting.
- **Dynamic required space:** Integrated speed-dependent required space $d_i(t) = a_i + b_i v_i(t)$.

Macroscopic Verification:

- **Convex curve recovery:** $b = 0.56$ s restores the correct fundamental-diagram shape.
- **Wave damping:** $b > 0$ suppresses artificial stop-and-go waves under constant spacing.
- **Parameter sensitivity:** baseline space a shifts congestion onset even with fixed b .

Theoretical Validation

The model reproduces the macroscopic trend while exposing how required-space calibration changes the curve.

The b -Parameter Discrepancy

While empirical studies measure human safety headways at $b \approx 1.06$ s, the numerical calibration matches the macroscopic fundamental diagram at $b \approx 0.56$ s.

Identifying the Limitation:

- **Model Short-Sightedness:** The 1D agents react *exclusively* to their immediate predecessor.
- Lacking reaction delays or down-the-line anticipation, they can safely maintain tighter gaps ($b = 0.56$ s) without causing system collapse.

Future Research Directions:

- **Cognitive & Behavioral Factors:** Incorporating human reaction delay times ($\Delta t_{\text{reaction}} > 0$) and cognitive distraction parameters.
- **Multi-Anticipative Models:** Expanding the interaction scope to look several steps ahead ($i + 2$, $i + 3$), mimicking real human proactive gaze.

THANK YOU!

Presented by Group 2